ÖZYEĞİN ÜNİVERSİTESİ

MATH 215
Probability and Random Variables

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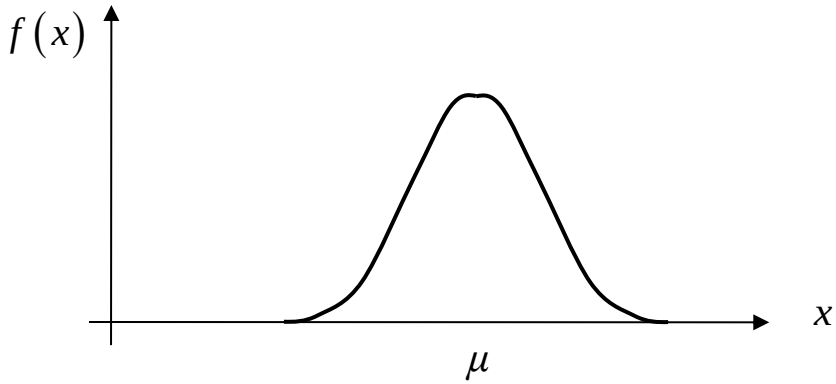
Parts of these notes are based on Murat Uysal's ECE316 lecture notes.

Normal (Gaussian) Random Variable

- X is a normal random variable with parameters μ and σ^2 if the pdf is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \qquad -\infty < x < \infty$$



- Common notation for normal random variable X : $X \sim N\left(\mu, \sigma^2\right)$
- $X \sim N(0,1)$ is said to be standard (unit) normal random variable.

Normal Random Variable (cont'd)

- The cdf of X

$$F(x) = P(X \le x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \frac{1}{\sigma} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$$

- A closed-form solution is not available. Making the variable change, we obtain a standard form which is available in software packages and also tabulated in textbooks

$$y = (t-\mu)/\sigma \qquad dy = dt/\sigma$$

$$t = -\infty \rightarrow y = -\infty$$

$$t = x \rightarrow y = (x-\mu)/\sigma$$

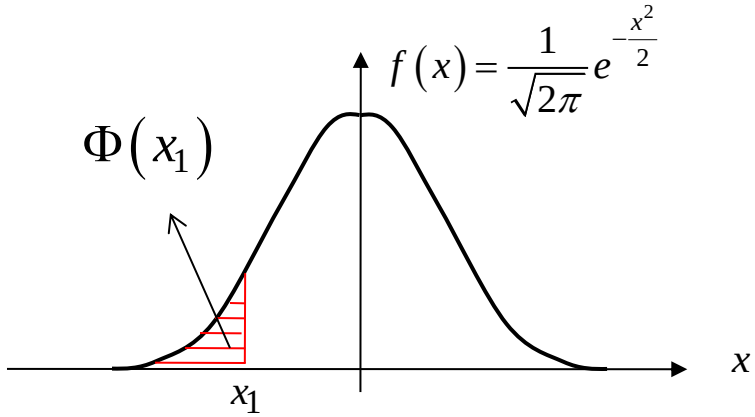
$$F(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-\frac{y^2}{2}} dy$$

Normal Random Variable (cont'd)

- For $X \sim N(0,1)$, the cdf is given by so-called Φ-function

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{y^2}{2}} dy$$

(Tabulated in the textbook for x>0)

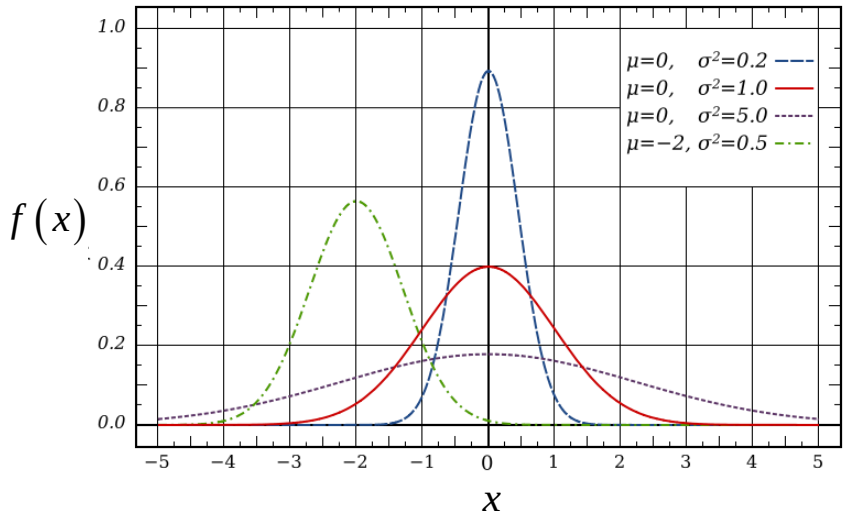


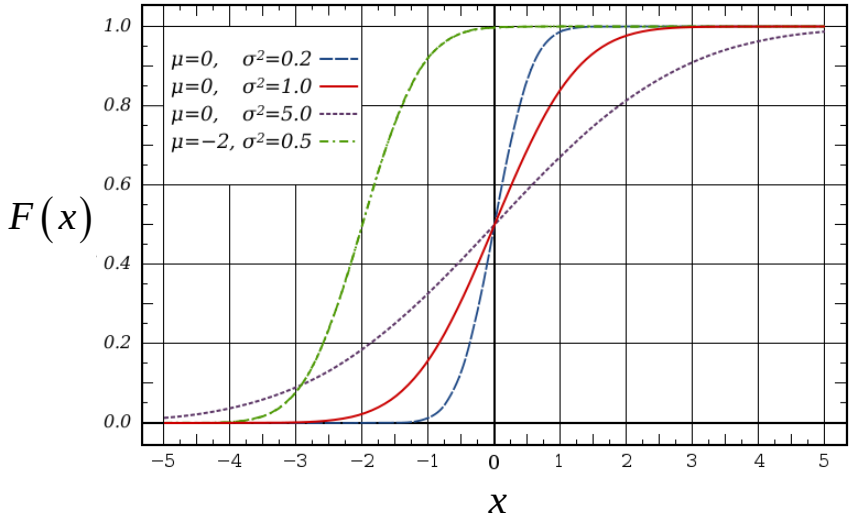
- For negative values, $\Phi(-x) = 1 - \Phi(x)$

- For $X \sim N(\mu, \sigma^2)$, the cdf can be written in terms of Φ-function

$$F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Normal Random Variable (cont'd)





Normal Random Variable (cont'd)

- Expected value

$$E(X) = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Making the variable change $t = (x-\mu)/\sigma \qquad dt = dx/\sigma$

$$E(X) = \int_{-\infty}^{\infty} (t\sigma + \mu) \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \frac{\sigma}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{\infty} t e^{-\frac{t^2}{2}} dt}_{=0 \text{ (see below)}} + \mu \underbrace{\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt}_{=1}$$

For the first integral, we introduce another variable change $u = t^2/2 \quad du = tdt$

$$\int_{-\infty}^{\infty} t e^{-\frac{t^2}{2}} dt = -e^{-u}\Big|_{\infty}^{\infty} = 0$$

$$\boxed{E(X) = \mu}$$

Normal Random Variable (cont'd)

- Variance

$$E\left(X^2\right) = \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Making the variable change $t = (x-\mu)/\sigma \qquad dt = dx/\sigma$

$$E\left(X^2\right) = \int_{-\infty}^{\infty} (\sigma t + \mu)^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$= \sigma^2 \underbrace{\int_{-\infty}^{\infty} t^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt}_{=1 \text{ (see next page)}} + \frac{2\sigma\mu}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{\infty} t e^{-\frac{t^2}{2}} dt}_{=0} + \mu^2 \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt}_{=1}$$

We will solve the first integral by "integration by parts"

Normal Random Variable (cont'd)

- Variance (cont'd)

$$u = t \qquad du = dt$$
$$dv = te^{-t^2/2} \qquad v = -e^{-t^2/2}$$

Integration by parts $\int u dv = uv - \int v du$

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-\frac{t^2}{2}} dt = \frac{1}{\sqrt{2\pi}} \left(\underbrace{-te^{-t^2/2} \Big|_{-\infty}^{\infty}}_{=0} + \int_{-\infty}^{\infty} e^{-t^2/2} dt \right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} dt = 1$$

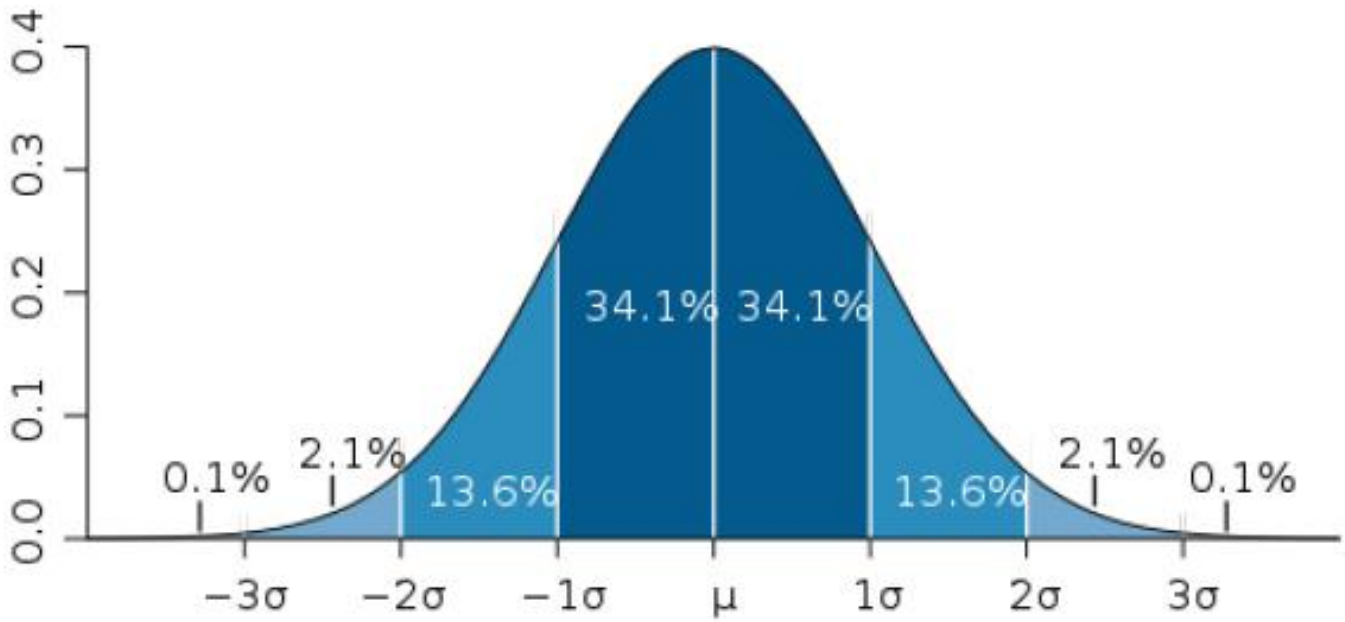
$$E\left(X^2\right) = \sigma^2 + \mu^2$$

$$\Rightarrow Var\left(X\right) = E\left(X^2\right) - E^2\left(X\right) = \sigma^2 + \mu^2 - \mu^2 = \sigma^2$$

$$\boxed{Var\left(X\right) = \sigma^2}$$

The Normal Distribution

- For any normal distribution:
  - About 68% of population is in $\mu \pm \sigma$.
  - About 95% of population is in $\mu \pm 2\sigma$.
  - About 99.7% of population is in $\mu \pm 3\sigma$.
- The above is usually referred to as the empirical rule. Its easier to understand with a picture:



The Normal Distribution (cont'd)

- **Problem:** Probabilities are found by calculating areas under the curve, but integrating the pdf for the normal distribution is tedious (and not possible with the "usual" methods).
- **Solution:** Computers! Its really easy to do this in Excel. For example, if $X \sim N(1,4)$ and you want $P(X \leq 0.5)$, you can just type
  - =NORM.DIST(0.5,1,2,1)
- **Problem:** So... how do I calculate probabilities for Quizzes/Exams?
- **Solution:** We'll use a table. Depending on what the problem wants, we'll manipulate the problem in the number on a table will give us our probabilities.
- **Problem:** There are an infinite number of (μ,σ^2) pairs. Are there an infinite number of normal probability tables?
- **Solution:** No. We'll translate every normal distribution into a Normal Distribution with $\mu = 0$; $\sigma^2 = 1$, called standardization by subtracting μ and dividing by σ. Note: This is called the standard normal distribution and usually expressed as Z.

The Normal Distribution (cont'd)

- On a test/quiz I will only provide the positive part of a normal table. How does this change your problem solving techniques?

- If your question is of the form $P(Z \leq z)$ where $z \geq 0$ then it changes nothing.

- If your question is of the form $P(Z \leq z)$ where $z \leq 0$ then you have to do an extra couple steps. Note that for $z \leq 0$, due to symmetry of the pdf of the normal distribution:
  - $P(Z \leq z) = P(Z \geq -z) = 1- P(Z \leq -z)$

Examples

- The manager of OzU cafeteria is planning for tomorrow's lunch. The chef has decided to produce two types of main dish: meatballs and vegetable beans. Each customer must select either one of the dishes (i.e., it is not allowed to select two dishes or no dishes). The manager ordered the kitchen to prepare 1500 servings of meatballs and 750 servings of vegetable beans for tomorrow.
  - Based on historical records, the manager estimates that the number of customers for tomorrow's lunch is normally distributed with mean 2000 and standard deviation 500.
  - Find the probability that some of the customers will be turned away as no main dish is available to purchase tomorrow.

Examples - 2

- A final examination at Ozyegin University is constructed so that the test scores are approximately normally distributed, with parameters μ and σ. The instructor assigns letter grades to the test scores as shown in the table below (this is the process of "grading on the curve"). What fraction of the class gets A, B, C, D, F?

Test Score	Letter grade
$\mu + \sigma < x$	A
$\mu < x < \mu + \sigma$	B
$\mu - \sigma < x < \mu$	C
$\mu - 2\sigma < x < \mu - \sigma$	D
$x < \mu - 2\sigma$	F

- A machine for filling cereal boxes has a standard deviation of 1 ounce of fill per box. What setting of the mean ounces of fill per box will allow 16-ounce boxes to overflow only 1% of the time? Assume that the ounces of fill per box are normally distributed.
- The times of first failure of a unit of a brand of ink jet printers are approximately normally distributed with a mean of 1500 hours and a standard deviation of 200 hours.

a. What fraction of these printers will fail before 1000 hours?

b. What should be the guarantee time for these printers if the manufacturer wants only 5% to fail within the guarantee period?

Normal Approximation to The Binomial Distribution

- Assume that X is a binomial random variable such that X~Binomial(N,β). Then we can use the following approximation when Nβ(1-β) is large (e.g., Nβ(1-β) ≥5:

$$P\left(a \leq \frac{X - N\beta}{\sqrt{N\beta(1-\beta)}} \leq b\right) = \Phi(b) - \Phi(a)$$

where $\Phi(x)$ is the cdf of standard normal distribution.

- **Example:** Let X~Binomial(40,0.5). What is P(X=20)?
  - The exact solution is $\binom{40}{20} 0.5^{40} = 0.1254$, which requires heavy computation.
  - Alternative method: Use normal approximation (notice 40(0.5)(0.5)>5):
  - $P(19.5 \leq X \leq 20.5) = P\left(\frac{19.5-20}{\sqrt{10}} \leq \frac{X-N\beta}{\sqrt{N\beta(1-\beta)}} \leq \frac{20.5-20}{\sqrt{10}}\right) = \Phi(0.16) - \Phi(-0.16) = 0.1272$

- Why use 19.5 and 20.5? Since binomial is a discrete distribution but normal is a continuous distribution, we use this trick (called continuity correction).

Examples

- The lifetimes of interactive computer chips produced by a certain semiconductor manufacturer are normally distributed with parameters $\mu = 1.4 \times 10^6$ hours and $\sigma = 3 \times 10^5$ hours. What is the approximate probability that a batch of 100 chips will contain at least 20 whose lifetimes are less than 1.8×10^6?

- Two types of coins are produced at a factory: a fair coin and a biased one that comes up heads 55 percent of the time. We have one of these coins, but do not know whether it is a fair coin or a biased one. In order to ascertain which type of coin we have, we shall perform the following statistical test: We shall toss the coin 1000 times. If the coin lands on heads 525 or more times, then we shall conclude that it is a biased coin, whereas if it lands on heads less than 525 times, then we shall conclude that it is a fair coin. If the coin is actually fair, what is the probability that we shall reach a false conclusion? What would it be if the coin were biased?